SHRESHTA RAAJ GUPTA

Computer Science Engineering Student | AI, Systems & Web3 Builder

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EXPERIENCE

Research Intern – Post-Quantum 5G Authentication

Indian Statistical Institute (ISI), Kolkata

2026 – Present | Kolkata

- Working on making 5G-AKA quantum-safe by integrating NIST post-quantum primitives (ML-KEM, ML-DSA) for constrained IoT devices with under 10 KB RAM, without changing the 3GPP architecture
- Analysing the Open5GS core-network source to locate the authentication and key-agreement paths where classical primitives can be replaced by PQC equivalents
- Part of a multi-institution effort with IIT Madras and TGS,

AI/ML Intern – Smart Pharmacy Inventory Prediction

IEEE EMBS Pune Chapter

June 2026 - July 2026 | Remote

- Built predictive-analytics models for pharmacy inventory management, forecasting medicine demand to reduce stock-out risk in a “Smart Pharmacy” system
- Completed as part of a one-month national online internship under IEEE EMBS Pune Chapter, IEEE Pune Section

IT and Operations Assistant

Ambrosia Restaurant (Family Business)

June 2017 – Present | Baranagar, Kolkata

- Maintained a LAN-based POS infrastructure connecting a central server with multiple distributed terminals including Kitchen KOT, Parcel Counter and AC Counter
- Diagnosed and resolved network and software failures in real time during peak operating hours
- Improved system reliability by proactively monitoring connectivity and addressing failure points under production load

PROJECTS

CRCS – Collusion-Resistant Credential System

Circom, Groth16, Poseidon, Zero-Knowledge Proofs

2026

- Contributed to a zero-knowledge credential system enabling selective disclosure of attributes without revealing underlying data
- Used Poseidon commitments and Circom/Groth16 circuits to prevent verifiers from colluding to link separate proofs to the same credential holder

Jigyasa – Voice-Enabled Hindi RAG

Python, FastAPI, FAISS, BM25, Cross-Encoder, Sarvam, Gemini

August 2026

TECHNICAL SKILLS

Python | TypeScript | Rust | Java | C++ | Solidity

Machine Learning | RAG | Information Retrieval | LLM Applications | Zero-Knowledge Proofs

FAISS | BM25 | LangChain | ChromaDB | scikit-learn | FastF1

React | Next.js | Node.js | FastAPI

Soroban | Smart Contracts | Docker | Git | Linux

ACHIEVEMENTS

4x Hackathon Winner
TCS Techday Hackathon, ISI Hackathon, Converge Ideathon and HexaFalls – Solana Track Winner and 3rd Runner-up

100+ LeetCode Problems
Strong foundations in DSA, OOP and DBMS

Stellar Inner Circle Member
Selected participant at Stellar Buildstation, Kolkata

IEEE Intern
IEEE EMBS Pune Chapter

EDUCATION

B.Tech, Computer Science and Engineering

Institute of Engineering and Management

2024 – 2028 | Salt Lake, Kolkata

LANGUAGES

English ●●●●●

Hindi ●●●●●

Bengali ●●●●◐

- Built a voice-first Hindi RAG assistant over a ∼7.77M-passage MS MARCO-XI corpus using Sarvam STT, hybrid dense+sparse retrieval, cross-encoder reranking and grounded generation
- Designed four length-routed chunking strategies, FAISS HNSW + BM25 retrieval with Reciprocal Rank Fusion, and an explicit abstention path for unsupported questions
- Evaluated on 1,000 held-out queries with Recall@10 of 0.637, NDCG@10 of 0.465 and MRR of 0.259; guardrails produced 0 false positives across 1,000 in-domain queries and 0 false negatives across 16 off-topic/unsafe cases
- Optimized and benchmarked the local pipeline at P50/P70/P100 latencies of 114.0/122.6/142.5 ms; a controlled K=10 configuration reduced latency by 44–48% without observed quality loss

Kairos – Intent-Based Non-Custodial Capital Delegation

TypeScript, Rust, Soroban, Next.js, Node.js

July 2026 – August 2026

- Built a non-custodial delegation system where an AI reasoning engine proposes trades while on-chain policies enforce spend limits, asset whitelists and time windows
- Designed a Soroban smart-wallet architecture with scoped, revocable delegations so funds remain under user custody
- Implemented a deterministic Market → Context → Memory → Strategy → Reasoning → Verification → Execution pipeline with only one LLM-dependent stage
- Shipped deployed Soroban contracts, a TypeScript SDK, Next.js dashboard and Node/Express backend with wallet-signature authentication

Ragged – Hybrid Retrieval-Augmented Generation Framework

Python, LangChain, ChromaDB, BM25, Ollama, Groq

June 2026 – Present

- Built a modular RAG framework combining BM25 sparse retrieval and dense vector search through Reciprocal Rank Fusion with cross-encoder reranking
- Added RAGAS-based evaluation for faithfulness, relevancy and correctness, with quality gates integrated into CI
- Designed a provider-agnostic LLM layer supporting local Ollama models and cloud providers without application-level changes

F1 Race Outcome Prediction

Python, pandas, scikit-learn, FastF1

December 2025 – Present

- Built an ML pipeline predicting F1 race classification using only pre-race features such as grid position, qualifying delta and constructor strength, with strict prevention of data leakage

Injective Token Sender

TypeScript, Injective SDK, Keplr, Cosmos SDK

2026

- Built a non-custodial dApp connecting Keplr wallet to Injective Testnet to read balances and send INJ through signed `MsgSend` transactions
- Handled Injective's Ethereum-derivation account model and precise 18-decimal base-unit amount handling
- Verified transaction success explicitly through on-chain result codes rather than assuming transaction inclusion implied success